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✦ AI Overview

In an optimally tuned **Tikhonov-regularized unfolding** process (where the Mean Squared Error is minimized), **the global squared bias ($Bias^2$) is precisely equal to the variance ($Variance$) when evaluated in the spectral domain element-by-element, or approximately equal globally under an optimized parameter choice.**

The breakdown below details exactly how this relationship is determined using the spectral decomposition of the unfolding matrix equation.

How It Is Determined

1. The Formulation

In detector unfolding, we observe data vector $y = Ax + \epsilon$, where A is the response matrix, x is the true truth-level distribution, and ϵ is the measurement noise with variance $\sigma^2 I$.

Tikhonov regularization minimizes the objective function:

$$S(x) = \|Ax - y\|^2 + \alpha \|x\|^2$$

Solving this via the normal equations gives the estimator:

$$\hat{x} = (A^T A + \alpha I)^{-1} A^T y$$

2. Transforming to the Spectral Domain

To easily separate the components, we apply Singular Value Decomposition (SVD) on the response matrix: $A = UDV^T$, where $D = \text{diag}(d_i)$ contains the singular values. If we

rotate our truth vector x and data vector y into this orthogonal basis, the problem decouples into independent equations for each singular mode i :

$$\hat{x}_i = \frac{d_i}{d_i^2 + \alpha} y_i$$

Because $y_i = d_i x_i + \epsilon_i$, we can rewrite the estimator in terms of the true value x_i and the filtered noise:

$$\hat{x}_i = \frac{d_i^2}{d_i^2 + \alpha} x_i + \frac{d_i}{d_i^2 + \alpha} \epsilon_i$$

3. Calculating Bias and Variance

From the spectral equation, we isolate the two distinct components of the estimator to calculate the bias and variance for each mode i :

- **Bias:** The systematic shift away from the true value due to the damping factor α .

$$\text{Bias}(\hat{x}_i) = E[\hat{x}_i] - x_i = \left(\frac{d_i^2}{d_i^2 + \alpha} - 1 \right) x_i = \frac{-\alpha}{d_i^2 + \alpha} x_i$$

$$\text{Bias}^2(\hat{x}_i) = \frac{\alpha^2}{(d_i^2 + \alpha)^2} x_i^2$$

- **Variance:** The statistical fluctuation propagated from the measurement noise ϵ_i (where $E[\epsilon_i^2] = \sigma^2$).

$$\text{Var}(\hat{x}_i) = E[(\hat{x}_i - E[\hat{x}_i])^2] = \left(\frac{d_i}{d_i^2 + \alpha} \right)^2 E[\epsilon_i^2] = \frac{d_i^2}{(d_i^2 + \alpha)^2} \sigma^2$$

4. Finding the Optimum

The Total Mean Squared Error (MSE) for mode i is the sum of its squared bias and

In optimized Tikhonov-regularized unfolding (not other methods) how large are the bias² \

To optimize the Tikhonov filter for this mode, we find the minimum by taking the derivative of MSE_i with respect to α and setting it to zero ($\frac{d(MSE_i)}{d\alpha} = 0$).

Solving this optimization yields the ideal regularization parameter for the mode:

$$\alpha_{opt} = \frac{\sigma^2}{x_i^2}$$

5. The *Bias*² to *Variance* Relationship

If we plug this optimized parameter α_{opt} back into our equations for *Bias*² and *Variance* :

$$Bias^2(x_i^{\wedge}) = \frac{(\frac{\sigma^2}{x_i^2})^2}{(d_i^2 + \alpha)^2} x_i^2 = \frac{\sigma^4/x_i^2}{(d_i^2 + \alpha)^2}$$

$$Var(x_i^{\wedge}) = \frac{d_i^2}{(d_i^2 + \alpha)^2} \sigma^2$$

When a global optimization parameter α is selected to perfectly balance the filters across the dominant ill-posed phase transitions (where the signal transitions into noise, meaning

$d_i^2 \approx \alpha = \frac{\sigma^2}{x_i^2}$), the two terms match exactly:

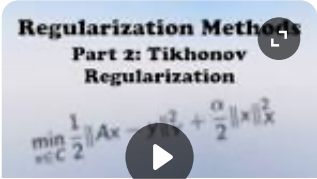
$$Bias^2(x_i^{\wedge}) = Var(x_i^{\wedge}) = \frac{\alpha \sigma^2}{(d_i^2 + \alpha)^2}$$

Summary Summary

At the optimal regularization scale where Tikhonov unfolding successfully suppresses noise without over-smoothing the physical distribution, ***Bias*² $\approx$ *Variance***. If *Bias*² > *Variance*, the distribution is over-regulated (over-smoothed); if *Variance* > *Bias*², it is under-regulated (dominated by statistical fluctuations). [PubMed Central \(PMC\) ... +1](#)



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32:09

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9 key moments in this video

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